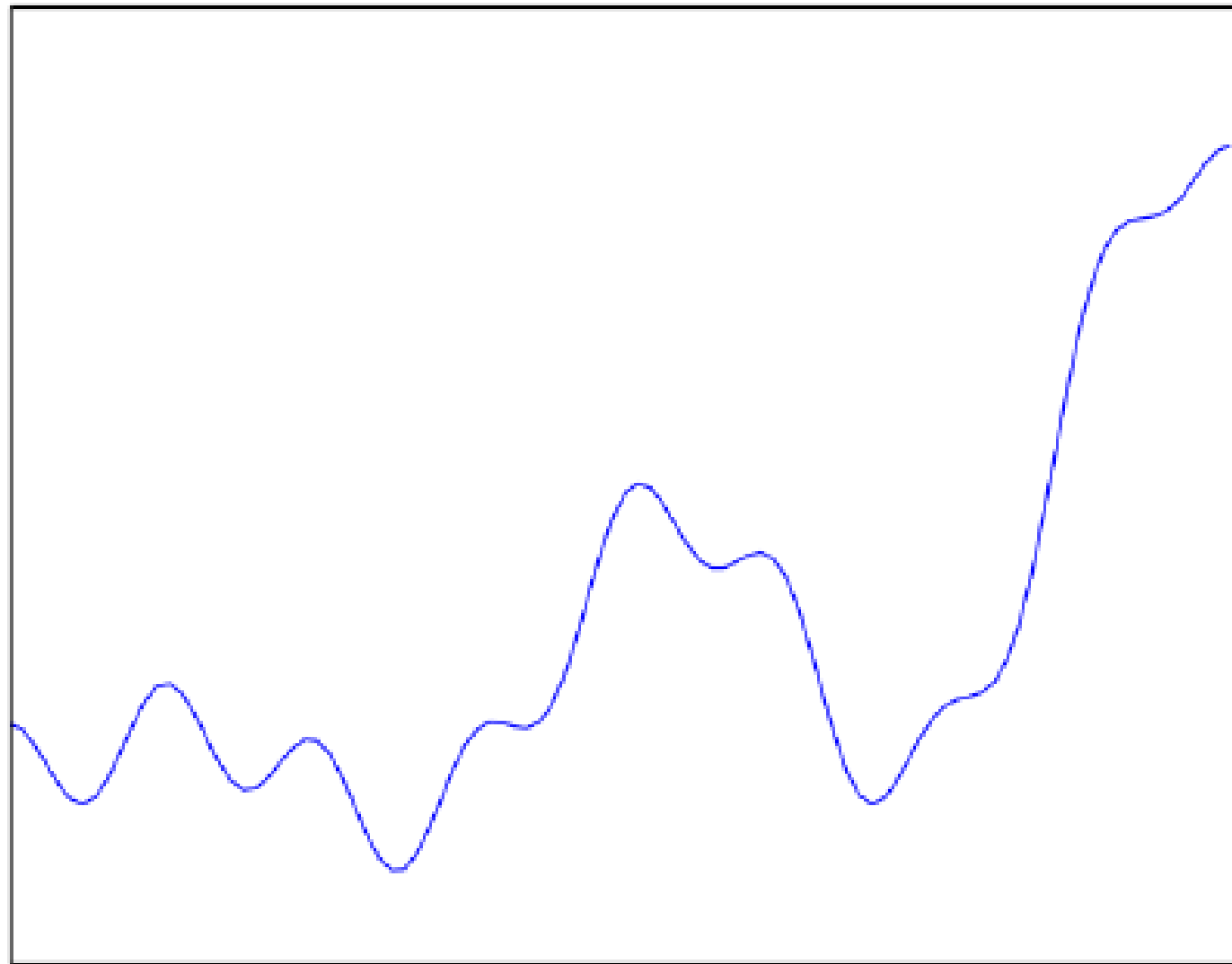




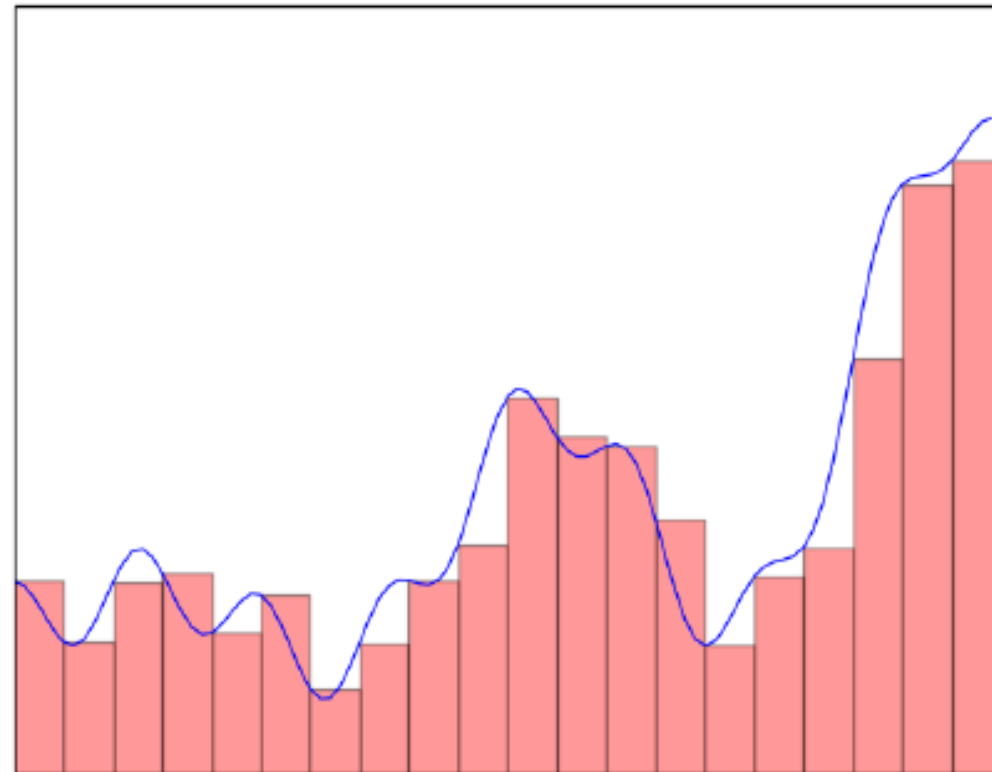
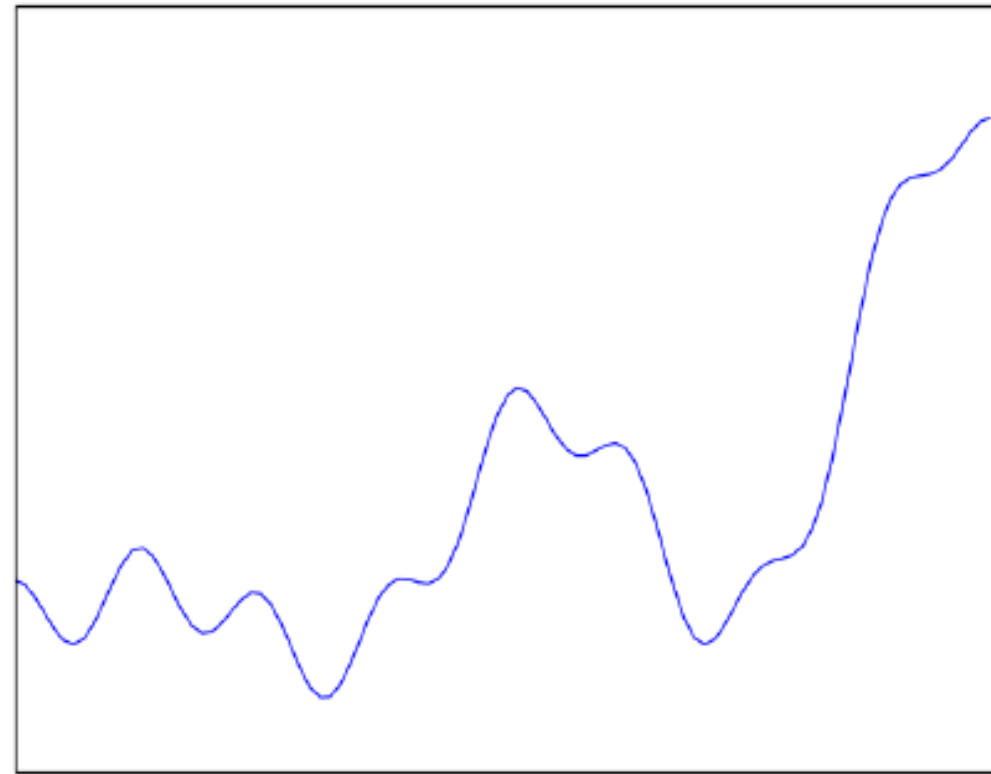
Arbitrary Function with Network of Neurons



We are interested in knowing whether a network of neurons can be used to represent an arbitrary function (like the one shown in the figure)



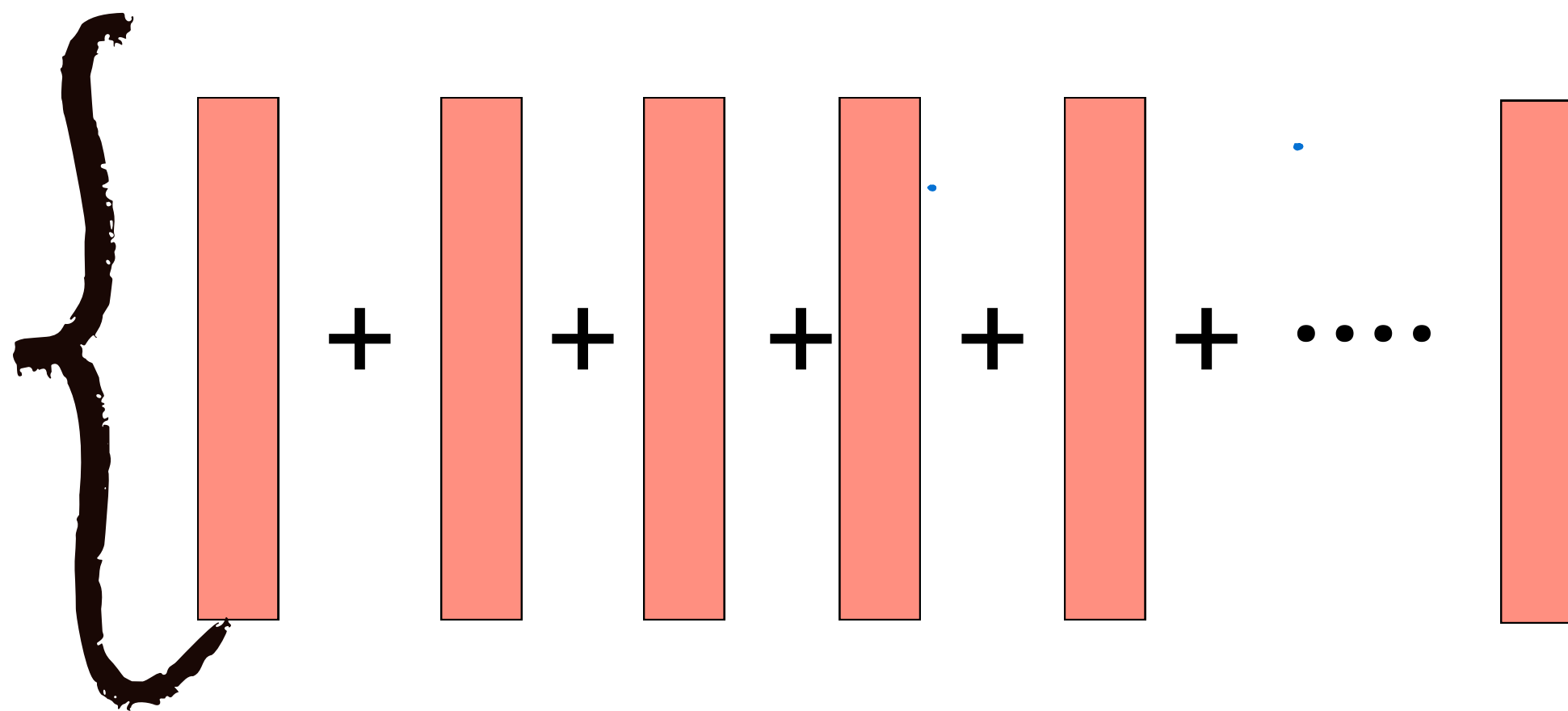
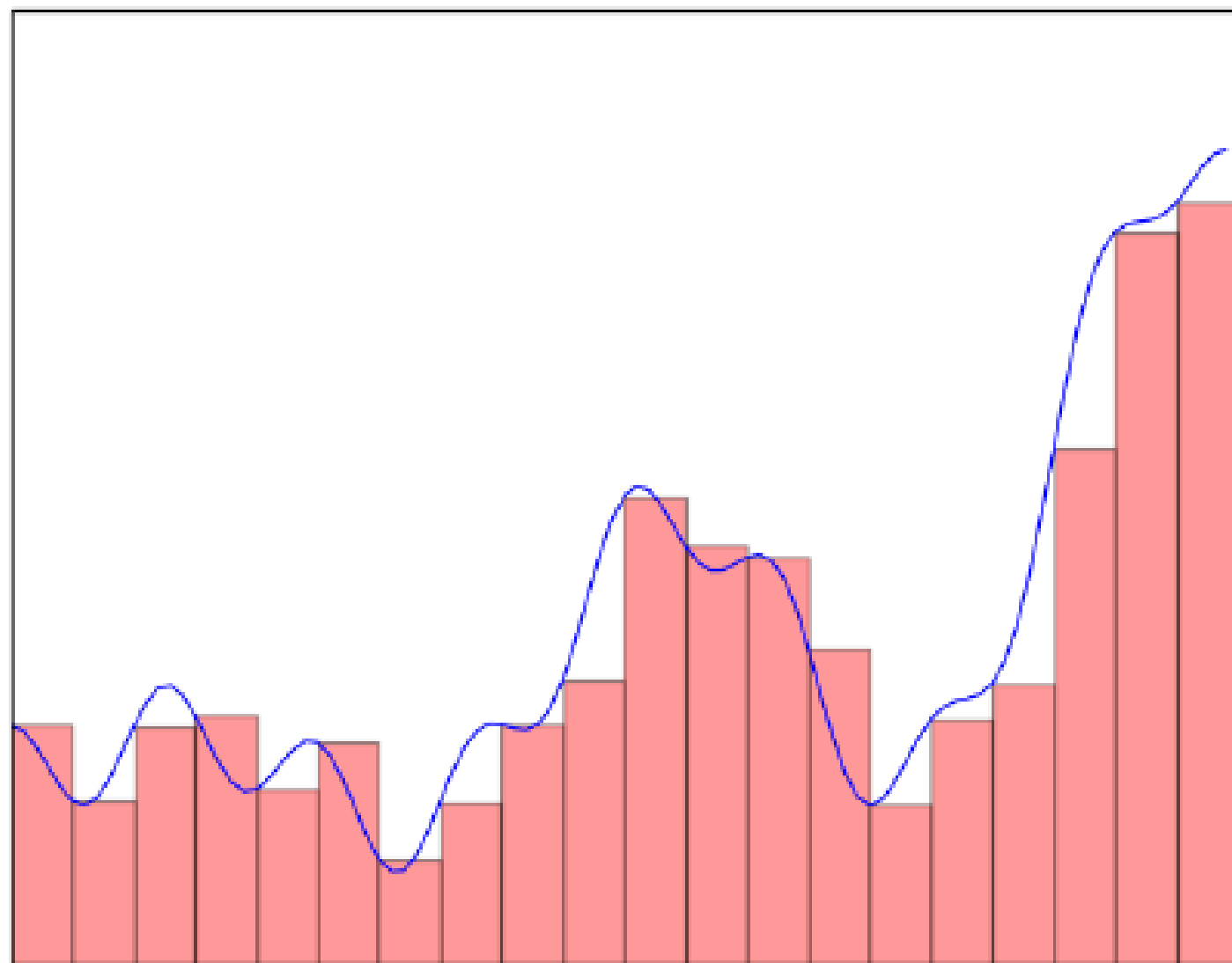
Arbitrary Function with Network of Neurons



- Our observation is : if we can have collection of vertical bars function it can approximate any arbitrary function
- If we have more such function, we can better approximate



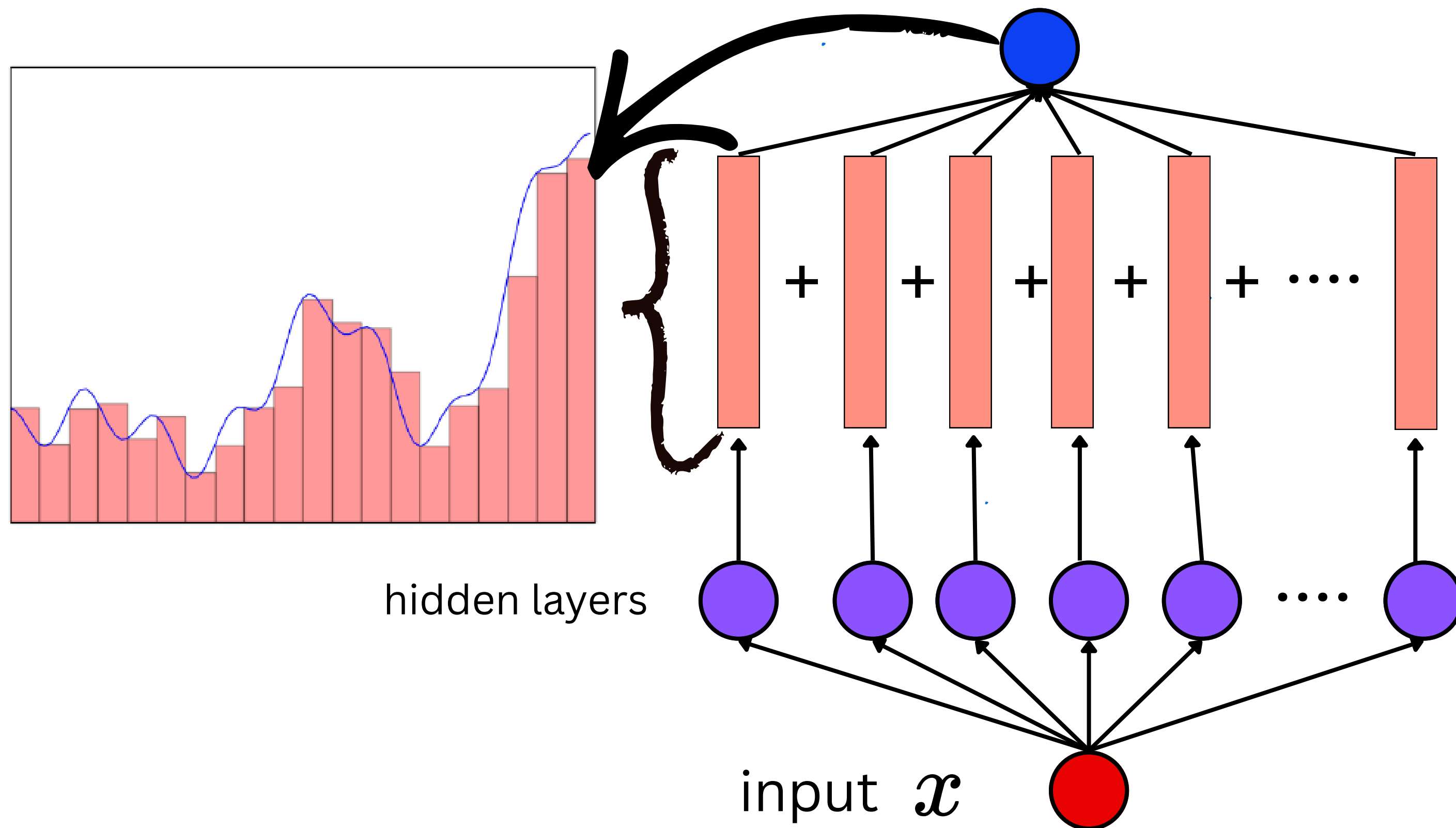
Arbitrary Function with Network of Neurons



The summation of such vertical bar function is the approximation of arbitrary function

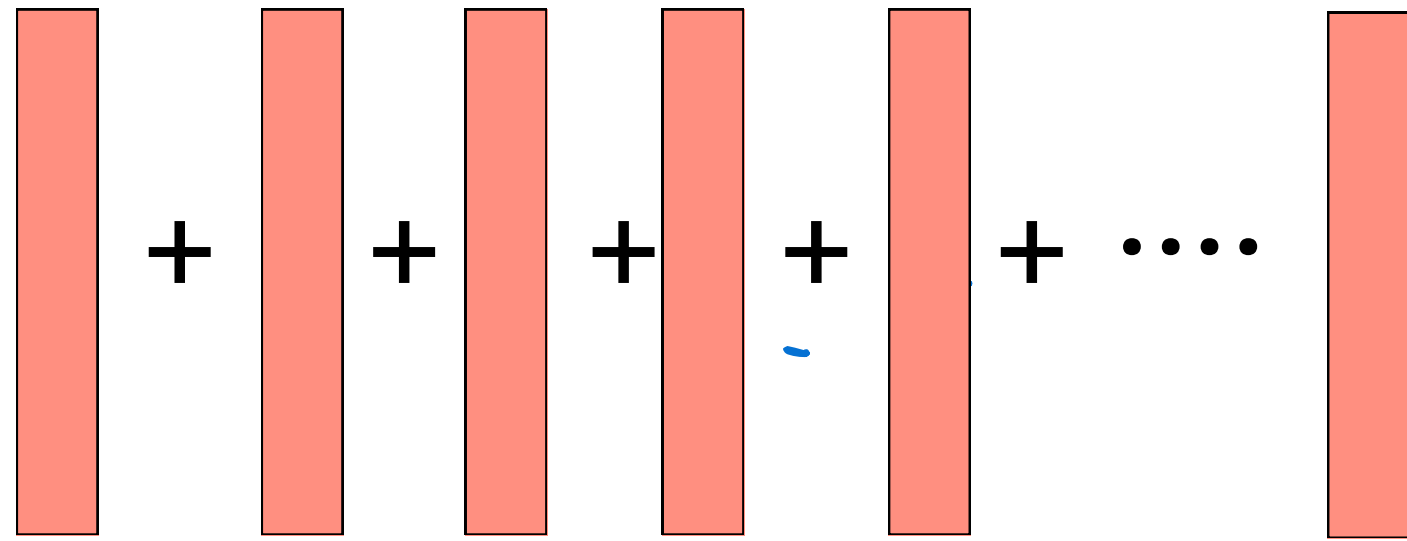


Arbitrary Function with Network of Neurons





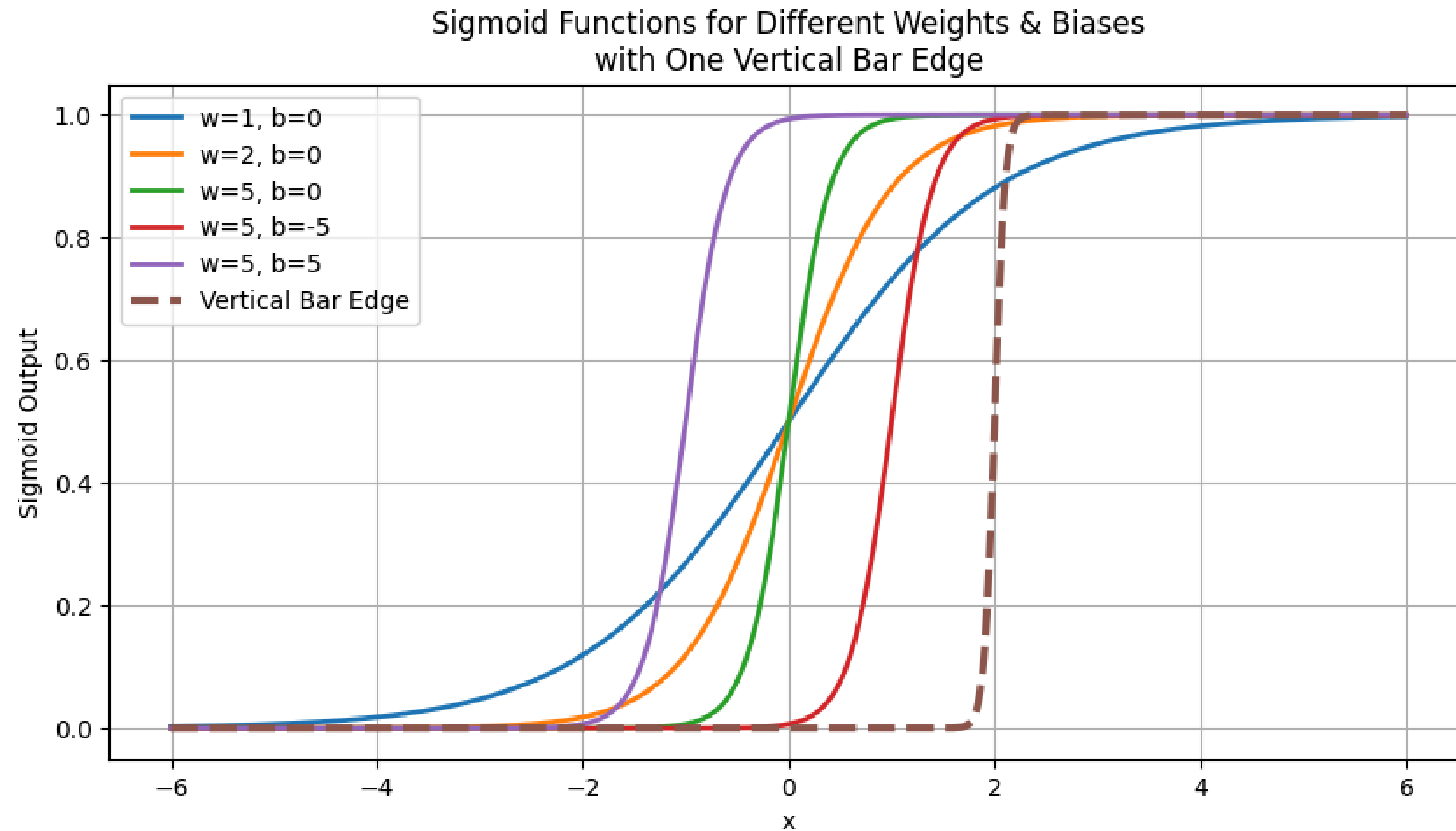
Arbitrary Function with Network of Neurons



As we have seen that summation of multiple bar can simulate any arbitrary function, now the question is : **how the bars can be formed?**

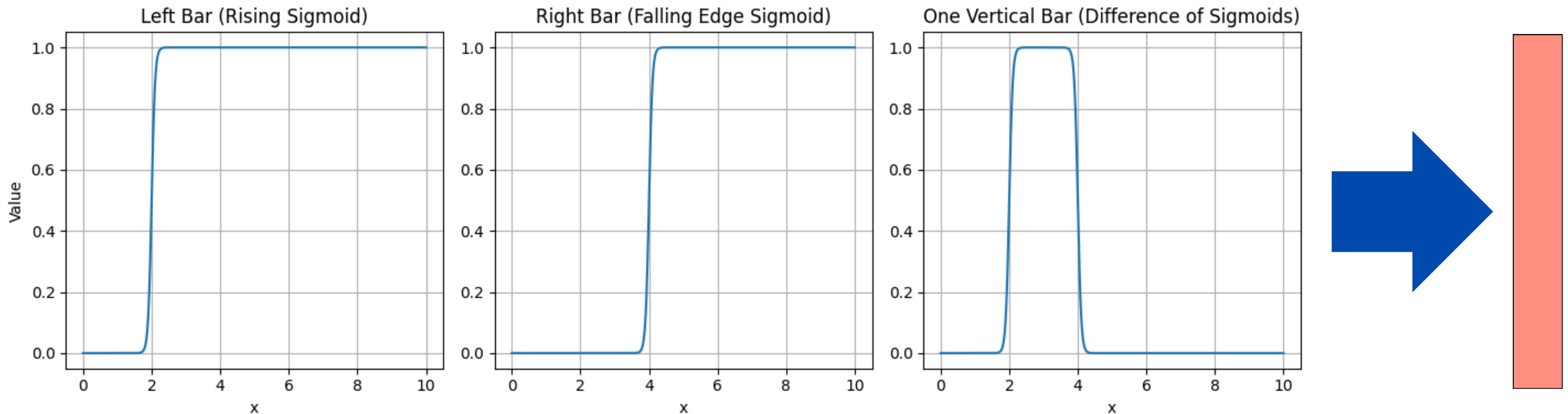


Arbitrary Function with Network of Neurons





Arbitrary Function with Network of Neurons



$$\text{bar}(x) = h \cdot [\sigma(k(x - a)) - \sigma(k(x - b))]$$

Where:

- $a, b \rightarrow$ left & right boundaries
- $h \rightarrow$ bar height
- $k \rightarrow$ steepness (larger = sharper bar)



Arbitrary Function with Network of Neurons

